Math 110 — Week 5

Quotient maps, isomorphism theorems, dual spaces (Notes Ch. 9–10, LADR 3E–3F, HW5)

§1 Facts used from Weeks 1–4

(W1) In a finite-dimensional space an independent list extends to a basis; n independent vectors, or n spanning vectors, in a space of dimension n form a basis; a subspace $U \subseteq V$ with $\dim U = \dim V < \infty$ equals V. A linear map can be defined by choosing its values on a basis, and two linear maps that agree on a basis are equal (Weeks 2, 3).

(W2) Rank–nullity: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$. T is injective $\iff$ $\operatorname{null} T = \{0\}$. The image $T(U)$ of a subspace U is a subspace (Week 3).

(W3) If $\dim V = \dim W < \infty$, then injective $\iff$ surjective $\iff$ isomorphism. Finite-dimensional spaces are isomorphic exactly when their dimensions are equal. An isomorphism and its inverse send bases to bases (Week 4).

(W4) $\dim \mathcal{L}(V, W) = \dim V \cdot \dim W$. The matrix of T: column j holds the coordinates of Tv_j. $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$ (Week 3).

(W5) Quotient space V/U: its elements are the sets $v+U$; $v+U = w+U \iff v-w \in U$; $(v+U)+(w+U) = (v+w)+U$, $\lambda(v+U) = \lambda v + U$. A formula that uses the name v must be checked to be *well-defined*: renaming the same set must not change the answer (Week 4).

(W6) A stochastic matrix P has entries ≥ 0 and columns summing to 1; P_{ij} is the probability of moving from state j to state i, and a distribution p becomes Pp after one step (Week 3).

(W7) If $p(a) = 0$ for a polynomial p, then $p = (x-a)q$ for a polynomial q (Week 4, W7).

§2 This week

§2.1 The quotient map and the dimension of a quotient (Ch. 9.1)

The question. In $V = \mathbb{R}^3$ let U be the line through $(1,1,1)$. The sets $v + U$ are the lines parallel to U. How many numbers are needed to say which line a vector lies on, and how do you pick a basis of V/U?

Definition (Def. 25 / LADR 3.104)**.** For a subspace $U \subseteq V$, the **quotient map** is $\pi\colon V \to V/U$, $\pi(v) = v + U$.

- π is linear: by the definition of the operations (W5),

$$\pi(\lambda v + w) = (\lambda v + w) + U = \lambda(v+U) + (w+U) = \lambda\pi(v) + \pi(w).$$

- π is surjective: every element of V/U is $v + U = \pi(v)$ for some v.
- $\operatorname{null} \pi = U$: $\pi(v)$ is the zero element $0 + U$ exactly when $v - 0 \in U$ (W5).

Proposition (Prop. 40 / LADR 3.105: dimension of a quotient)
If V is finite-dimensional, $\dim(V/U) = \dim V - \dim U$.

Proof. 1. The quotient map $\pi\colon V \to V/U$ is linear, surjective, and $\operatorname{null}\pi = U$ (bullets above).

2. Rank–nullity (W2) for π: $\dim V = \dim\operatorname{null}\pi + \dim\operatorname{range}\pi = \dim U + \dim(V/U)$.

□

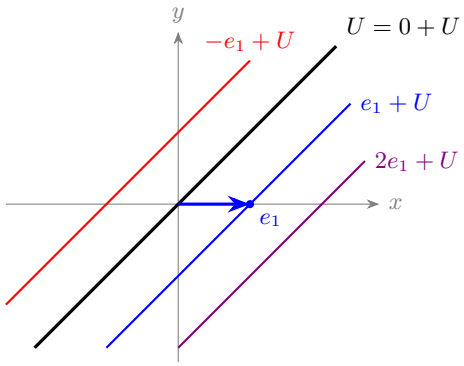
Proposition (A basis of V/U)
Let V be finite-dimensional, $u_1, \dots, u_k$ a basis of U, and $u_1, \dots, u_k, x_1, \dots, x_r$ a basis of V. Then $x_1+U, \dots, x_r+U$ is a basis of V/U.

Proof. 1. π is surjective, so it sends the basis $u_1, \dots, u_k, x_1, \dots, x_r$ of V to a spanning list of V/U: every $v + U$ is $\pi(\sum a_i u_i + \sum b_j x_j) = \sum a_i \pi(u_i) + \sum b_j \pi(x_j)$.

2. $\pi(u_i) = 0$, so dropping them leaves the spanning list $x_1 + U, \dots, x_r + U$.

3. It has $r = \dim V - \dim U = \dim(V/U)$ elements (Prop. 40), and a spanning list of length dim is a basis (W1).

□



$V = \mathbb{R}^2$, $U = \operatorname{span}((1,1))$. Each line parallel to U is one element of V/U, and these elements are exactly the multiples $c\,(e_1 + U) = ce_1 + U$. So $e_1 + U$ is a basis of V/U: here U has basis $(1,1)$, which extends to the basis $(1,1), e_1$ of $\mathbb{R}^2$.

The question answered. $(1,1,1), e_2, e_3$ is a basis of $\mathbb{R}^3$, so $e_2 + U$, $e_3 + U$ is a basis of $\mathbb{R}^3/U$ and $\dim(\mathbb{R}^3/U) = 3 - 1 = 2$. Two numbers say which line a vector lies on: $(x,y,z) = x(1,1,1) + (y-x)e_2 + (z-x)e_3$, so $(x,y,z) + U = (y-x)(e_2 + U) + (z-x)(e_3 + U)$.

Shortcuts

- **Basis of V/U.** Extend a basis of U to a basis of V; the added vectors $+U$ are a basis, and $\dim V/U = \dim V - \dim U$. Reason: the two propositions above. Condition: V finite-dimensional. Instance: for $U = \operatorname{span}((1,1,0)) \subseteq \mathbb{R}^3$, the list $(1,1,0), e_2, e_3$ is a basis of $\mathbb{R}^3$, so $e_2 + U$, $e_3 + U$ is a basis of $\mathbb{R}^3/U$.

Traps

- Elements of V/U are sets. In $\mathbb{R}^2/\operatorname{span}(e_2)$, $e_1 + U$ and $(e_1 + 5e_2) + U$ are the same element, so "the basis vector e_1" of V/U does not make sense; the basis element is the set $e_1 + U$.

§2.2 The first isomorphism theorem (Ch. 9.1)

The question. $T\colon \mathbb{R}^3 \to \mathbb{R}^2$, $T(x,y,z) = (x+y,\, x+y)$, sends $(1,0,0)$ and $(0,1,0)$ to the same vector and misses $(1,0)$. Can we change the domain and the target so that the same formula becomes a bijection?

Theorem (Prop. 41 / LADR 3.107: first isomorphism theorem)
Let $T \in \mathcal{L}(V,W)$. The formula

$$\widetilde{T}\colon V/\operatorname{null} T \to \operatorname{range} T, \qquad \widetilde{T}(v + \operatorname{null} T) = Tv$$

defines an isomorphism. Moreover $T = \iota \circ \widetilde{T} \circ \pi$, where $\pi\colon V \to V/\operatorname{null} T$ is the quotient map and $\iota\colon \operatorname{range} T \to W$, $\iota(w) = w$, is the inclusion.

The idea

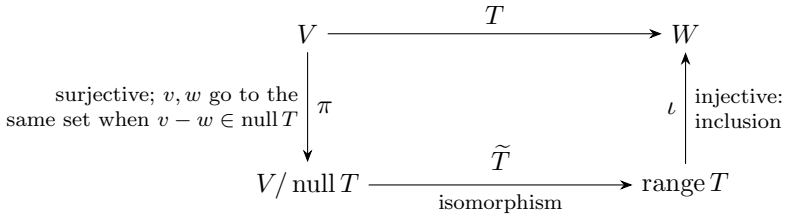
Surjectivity. Replacing the target W by $\operatorname{range} T$ makes the map surjective.

Injectivity. Passing to the quotient by $\operatorname{null} T$ makes the map injective. Two vectors have the same image exactly when their difference is in $\operatorname{null} T$:

$$Tv = Tw \iff T(v-w) = 0 \iff v - w \in \operatorname{null} T,$$

and $v - w \in \operatorname{null} T$ is exactly the condition for v and w to give the same element of $V/\operatorname{null} T$ (W5). So the pairs of vectors with the same image under T are exactly the pairs that give the same element of $V/\operatorname{null} T$, and on $V/\operatorname{null} T$ different elements have different images.

Read left to right, the chain says the formula $\widetilde{T}(v + \operatorname{null} T) = Tv$ is well-defined (same set $\Rightarrow$ same value). Read right to left it says $\widetilde{T}$ is injective (same value $\Rightarrow$ same set).



Every linear map is a quotient map (surjective), then an isomorphism, then an inclusion (injective). Going down, across and up gives the same result as T.

Proof. 1. **Well-defined** (the chain read left to right). Suppose $v + \operatorname{null} T = w + \operatorname{null} T$. Then $v - w \in \operatorname{null} T$ (W5), so $Tv - Tw = T(v-w) = 0$, so $Tv = Tw$. The value $\widetilde{T}(v + \operatorname{null} T)$ does not depend on the name v.

2. **Linear.** For $v, w \in V$ and $\lambda \in \mathbb{F}$,
$\widetilde{T}\big(\lambda(v+\operatorname{null} T) + (w+\operatorname{null} T)\big) = \widetilde{T}\big((\lambda v + w) + \operatorname{null} T\big) = T(\lambda v + w) = \lambda Tv + Tw = \lambda\widetilde{T}(v+\operatorname{null} T) + \widetilde{T}(w+\operatorname{null} T)$.

3. **Injective** (the chain read right to left). Suppose $\widetilde{T}(v + \operatorname{null} T) = 0$. Then $Tv = 0$, so $v \in \operatorname{null} T$, so $v + \operatorname{null} T = 0 + \operatorname{null} T$ is the zero element of the quotient.

4. **Surjective onto** $\operatorname{range} T$**.** Every element of $\operatorname{range} T$ is $Tv = \widetilde{T}(v + \operatorname{null} T)$ for some v.

5. **Factorisation.** For every $v \in V$,
$$\iota(\widetilde{T}(\pi(v))) = \iota(\widetilde{T}(v + \operatorname{null} T)) = \iota(Tv) = Tv.$$

□

Dimensions:

$$\dim V - \dim \operatorname{null} T = \dim(V/\operatorname{null} T) = \dim \operatorname{range} T.$$

This is rank–nullity again (not a new proof, since Prop. 40 used rank–nullity).

Example 43: differentiation. $D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $Dp = p'$, is surjective: every polynomial $q = \sum a_k x^k$ is the derivative of $\sum \frac{a_k}{k+1} x^{k+1}$. Its null space is the constants $\mathbb{R}$. The induced isomorphism is

$$\mathcal{P}(\mathbb{R})/\mathbb{R} \to \mathcal{P}(\mathbb{R}), \qquad p + \mathbb{R} \mapsto p'.$$

Its inverse is the indefinite integral $q \mapsto \int q\,dx$, whose value is a polynomial "$+C$", that is, an element of $\mathcal{P}(\mathbb{R})/\mathbb{R}$.

Infinite-dimensional V: the quotient can have either kind of dimension. In $V = \mathcal{P}(\mathbb{R})$ let $U = \{p : p(0) = 0\}$. The map $p \mapsto p(0)$ is linear, onto $\mathbb{R}$ (the constant c goes to c), with null space U. By Prop. 41, $V/U \cong \mathbb{R}$ has dimension 1. For $U = \{0\}$ instead, $V/U \cong V$ is infinite-dimensional: the quotient map $v \mapsto v + \{0\}$ is surjective with null space $\{0\}$, so it is an isomorphism.

Remark from the notes: a subspace U comes with an injective map (its inclusion), and every injective map $i\colon U \to V$ identifies U with the subspace $i(U)$. In the same way a quotient V/U comes with a surjective map π, and every surjective $T\colon V \to W$ identifies W with the quotient $V/\operatorname{null} T$.

The question answered. Yes. The null space and range are

$$\operatorname{null} T = \{(x,y,z) : x + y = 0\} = \operatorname{span}((1,-1,0),(0,0,1)), \qquad \operatorname{range} T = \operatorname{span}((1,1)).$$

By Prop. 41, $(x,y,z) + \operatorname{null} T \mapsto (x+y, x+y)$ is an isomorphism $\mathbb{R}^3/\operatorname{null} T \to \operatorname{span}((1,1))$; both have dimension 1.

Shortcuts

- **Proving "quotient $\cong X$".** Write a surjective linear map from the numerator onto X, find its null space, and quote the first isomorphism theorem; well-definedness is then part of the theorem. Reason: Prop. 41. Instance: $\mathbb{R}^3/\operatorname{span}(e_3) \cong \mathbb{R}^2$, since $(x,y,z) \mapsto (x,y)$ is onto $\mathbb{R}^2$ with null space $\operatorname{span}(e_3)$.
- **Basis of $V/\operatorname{null} T$ when T is onto $\mathbb{F}^k$.** Pick $p_1, \dots, p_k$ with $Tp_i = e_i$; then $p_1 + \operatorname{null} T, \dots, p_k + \operatorname{null} T$ is a basis. Reason: $\widetilde{T}$ is an isomorphism sending $p_i + \operatorname{null} T$ to e_i, and the inverse of an isomorphism sends a basis to a basis (W3). Instance: HW5 P1, with $p_1 = 1 - x$, $p_2 = x$.

Traps

- $\dim V/U = \dim V - \dim U$ needs V finite-dimensional: $\mathcal{P}(\mathbb{R})/\{p : p(0) = 0\}$ has dimension 1, although $\mathcal{P}(\mathbb{R})$ and $\{p : p(0) = 0\}$ are both infinite-dimensional.
- The target of $\widetilde{T}$ is $\operatorname{range} T$, not W: in the question, $\mathbb{R}^3/\operatorname{null} T$ has dimension 1 and is not isomorphic to $\mathbb{R}^2$.

§2.3 Induced maps; second and third isomorphism theorems (Ch. 9.2–9.3)

The question. Differentiation D on $\mathcal{P}(\mathbb{R})$ maps the constants to 0. Does $p + \mathbb{R} \mapsto p'$ give a well-defined map on $\mathcal{P}(\mathbb{R})/\mathbb{R}$? Does $p + \mathbb{R} \mapsto p(0)$? In general, when does a formula $v + U_1 \mapsto Tv + U_2$ make sense?

Proposition (Prop. 42: when a map passes to quotients)

Let $T\colon V \to W$ be linear, $U_1 \subseteq V$ and $U_2 \subseteq W$ subspaces with $T(U_1) \subseteq U_2$. Then $\widetilde{T}\colon V/U_1 \to W/U_2$, $\widetilde{T}(v + U_1) = Tv + U_2$, is a well-defined linear map.

The idea

The formula uses the name v, so the only thing to prove is that renaming does not matter. Two names v, w of the same set differ by an element of U_1. Their images Tv, Tw then differ by an element of $T(U_1)$. For the outputs to be the same set we need that difference to be in U_2. So the hypothesis $T(U_1) \subseteq U_2$ is what is required. It is also necessary: if $T(U_1) \not\subseteq U_2$, pick $u \in U_1$ with $Tu \notin U_2$; then u and 0 name the same element of V/U_1, but $Tu + U_2 \neq 0 + U_2$.

Proof. 1. **Well-defined.** Suppose $v + U_1 = w + U_1$. Then $v - w \in U_1$, so $Tv - Tw = T(v - w) \in T(U_1) \subseteq U_2$, so $Tv + U_2 = Tw + U_2$.

2. **Linear.** For $v, w \in V$ and $\lambda \in \mathbb{F}$,

$$\widetilde{T}(\lambda(v + U_1) + (w + U_1)) = T(\lambda v + w) + U_2 = \lambda(Tv + U_2) + (Tw + U_2).$$

□

Special case $U_2 = \{0\}$ (HW5 P2). In $W/\{0\}$ each element $w + \{0\}$ is the one-element set $\{w\}$, and the map $w \mapsto w + \{0\}$ is an isomorphism $W \to W/\{0\}$ (it is the quotient map, surjective, with null space $\{0\}$). So if $U \subseteq \operatorname{null} T$, then T induces $\widetilde{T}\colon V/U \to W$, $\widetilde{T}(v + U) = Tv$. The first isomorphism theorem is the case $U = \operatorname{null} T$.

Theorem (Prop. 43: second isomorphism theorem)

For subspaces $U, W \subseteq V$ there is a natural isomorphism

$$\varphi\colon U/(U \cap W) \to (U + W)/W, \qquad \varphi\big(u + (U \cap W)\big) = u + W.$$

"Natural" in the notes means: the isomorphism is given by a formula that uses only the subspaces themselves, with no choice of basis or complement.

The idea

The plan for every statement "quotient $\cong$ something" is the same: find a surjective linear map from the numerator onto the something, compute its null space, and apply the first isomorphism theorem. Here the numerator is U and the target is $(U + W)/W$, and the map is $u \mapsto u + W$.

In finite dimensions the theorem gives $\dim U - \dim(U \cap W) = \dim(U + W) - \dim W$, which is the formula for $\dim(U + W)$ from Week 2.

Proof. 1. Define $S\colon U \to (U + W)/W$ by $S(u) = u + W$. It is linear, because it is the quotient map of $U + W$ restricted to U.

2. S **is surjective.** An element of $(U + W)/W$ is $(u + w) + W$ with $u \in U$, $w \in W$. Since $(u + w) - u = w \in W$, this equals $u + W = S(u)$.

3. $\operatorname{null} S = U \cap W$**.** For $u \in U$: $S(u) = 0 + W \iff u \in W \iff u \in U \cap W$.

4. By the first isomorphism theorem, $U/\operatorname{null} S \to \operatorname{range} S$, that is, $U/(U \cap W) \to (U + W)/W$, $u + (U \cap W) \mapsto u + W$, is an isomorphism.

□

Theorem (Prop. 44: third isomorphism theorem)
Let $W \subseteq U \subseteq V$ be subspaces. Then $U/W = \{u + W : u \in U\}$ is a subspace of V/W, and

$$\varphi\colon (V/W)/(U/W) \to V/U, \qquad \varphi\big((v + W) + U/W\big) = v + U,$$

is a well-defined isomorphism.

The plan is the same as for Prop. 43: find a surjective map from V/W onto V/U and compute its null space. The candidate is $v + W \mapsto v + U$, which replaces the set $v + W$ by the larger set $v + U$.

Proof. 1. U/W **is a subspace.** U/W is the image of the subspace U under the quotient map $V \to V/W$, and the image of a subspace under a linear map is a subspace (W2).

2. The identity map of V sends W into U. By Prop. 42 it induces the linear map $S\colon V/W \to V/U$, $S(v+W) = v+U$.

3. S **is surjective**: every $v + U$ is $S(v + W)$.

4. $\operatorname{null} S = U/W$: $S(v + W) = 0 + U \iff v \in U \iff v + W \in U/W$. (For the last step: if $v + W = u + W$ with $u \in U$, then $v = u + (v - u) \in U + W = U$.)

5. The first isomorphism theorem gives the isomorphism $(V/W)/(U/W) \to V/U$, $(v+W)+U/W \mapsto S(v+W) = v+U$. □

- Dimension check: $(\dim V - \dim W) - (\dim U - \dim W) = \dim V - \dim U$.
- Use in induction proofs: if $W \neq \{0\}$, then V/W has smaller dimension than V.

The question answered. $p + \mathbb{R} \mapsto p'$ is well defined, since $D(\mathbb{R}) = \{0\}$; it is Ex. 43. $p + \mathbb{R} \mapsto p(0)$ is not: x and $x + 1$ name the same element of $\mathcal{P}(\mathbb{R})/\mathbb{R}$ but give 0 and 1. In general, $v + U_1 \mapsto Tv + U_2$ makes sense exactly when $T(U_1) \subseteq U_2$ (Prop. 42 and its idea box).

Shortcuts

- **Defining a map out of** V/U**.** The formula $v + U \mapsto Tv$ is valid exactly when $U \subseteq \operatorname{null} T$. Reason: Prop. 42 with $U_2 = \{0\}$. Instance: $(x, y, z) + \operatorname{span}(e_3) \mapsto x + 2y$ is well defined, since $(x, y, z) \mapsto x + 2y$ is 0 on e_3; $(x, y, z) + \operatorname{span}(e_3) \mapsto z$ is not, since e_3 and 0 name the same set but give 1 and 0.

Traps

- The second isomorphism theorem has $U \cap W$ in the first denominator, not W: U/W is not defined unless $W \subseteq U$. In $\mathbb{R}^3$ with $U = \operatorname{span}(e_1, e_2)$, $W = \operatorname{span}(e_2, e_3)$, the theorem says $U/\operatorname{span}(e_2) \cong \mathbb{R}^3/W$; both have dimension 1.

§2.4 Dual space and dual basis (Ch. 10.1)

The question. A vector can be studied through numbers that depend linearly on it: a coordinate, the value of a polynomial at a point, an integral. What is the space of all such functions, and what is a basis of it when $V = \mathbb{R}^2$ with the basis $(1, 1), (1, -1)$?

Definition (Def. 26, 27 / LADR 3.108, 3.112)**.** A **linear functional** on V is a linear map $\varphi\colon V \to \mathbb{F}$. The **dual space** is $V' = \mathcal{L}(V, \mathbb{F})$. If $v_1, \ldots, v_n$ is a basis of V, its **dual basis** is the list $\varphi_1, \ldots, \varphi_n \in V'$ with

$$\varphi_j(v_k) = \delta_{jk} = \begin{cases} 1 & j = k \\ 0 & j \neq k. \end{cases}$$

- Elements of V' are also called linear forms, one-forms or covectors; these are different names for the same objects.

- V' has nothing to do with derivatives. Other books (and the practice midterm) write V^*. The notes later use the asterisk for a different operation, so in this course write V'.
- Examples on $\mathcal{P}(\mathbb{R})$ (LADR 3.109): $p \mapsto p(4)$, $p \mapsto 3p''(5) + 7p(4)$, $p \mapsto \int_0^1 p(x)\,dx$. Each is linear because evaluation, differentiation and integration are.
- φ_j exists and is unique because a linear map is determined by its values on a basis (W1).
- **What φ_j does:** $\varphi_j(a_1v_1 + \cdots + a_nv_n) = a_j$. It reads off the j-th coordinate.
- Ex. 44: for $\mathbb{F}^n$ (columns) a functional is a row vector, acting by row times column. The dual basis of $e_1, \ldots, e_n$ is the list of rows $(1\ 0\ \cdots\ 0), \ldots, (0\ \cdots\ 0\ 1)$.
- On $\mathbb{R}^n$ the dot product gives a map $v \mapsto (w \mapsto w \cdot v)$, which sends a column v to the row v^{T}. It is an isomorphism $\mathbb{R}^n \to (\mathbb{R}^n)'$, but it depends on the dot product: a different inner product gives a different map. The double-dual map in the next section uses nothing of this kind.
- $\dim V' = \dim V \cdot \dim \mathbb{F} = \dim V$ (W4). So $V' \cong V$ when V is finite-dimensional. But an isomorphism has to be chosen, for example $v_j \mapsto \varphi_j$, and a different basis gives a different one. For $V = \mathbb{R}^2$: the basis e_1, e_2 gives $e_1 \mapsto (1\ 0)$; the basis $e_1, (1,1)$ has the dual basis $(1\ -1), (0\ 1)$, so it gives $e_1 \mapsto (1\ -1)$. The two isomorphisms send e_1 to different rows.

Proposition (Prop. 45 / LADR 3.116: the dual basis is a basis)
If $v_1, \ldots, v_n$ is a basis of V, its dual basis $\varphi_1, \ldots, \varphi_n$ is a basis of V'.

The idea

The technique for functionals: *to get information out of an equation between functionals, evaluate both sides on basis vectors.* Evaluating $\lambda_1\varphi_1 + \cdots + \lambda_n\varphi_n$ at v_k makes every term vanish except the k-th, and returns λ_k.

Proof. 1. Suppose $\lambda_1\varphi_1 + \cdots + \lambda_n\varphi_n = 0$ (the zero functional). Evaluate at v_k:

$$0 = \lambda_1\varphi_1(v_k) + \cdots + \lambda_n\varphi_n(v_k) = \lambda_k.$$

This holds for every k, so $\lambda_1 = \cdots = \lambda_n = 0$: the list is independent.

2. It has $n = \dim V'$ vectors, so it is a basis (W1).

□

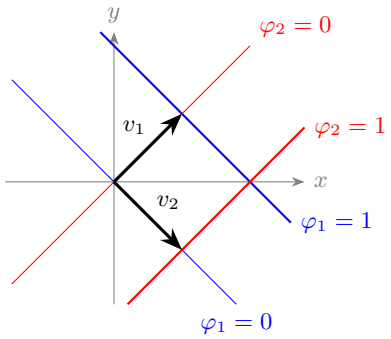
Coordinates. Of a vector: $v = \sum_j \varphi_j(v)\, v_j$. Of a functional: $\psi = \sum_j \psi(v_j)\, \varphi_j$ (HW5 P4). The coefficient of φ_j is found by evaluating ψ at v_j.

Dual basis of $(1,1), (1,-1)$ in $\mathbb{R}^2$.

1. Put the basis vectors as the columns of M. Row j of M^{-1} times column k of M is entry (j,k) of $M^{-1}M = I$, which is δ_{jk}. So the rows of M^{-1} are the dual basis.

$$M = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad M^{-1} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}.$$

2. So $\varphi_1(x,y) = \frac{1}{2}(x+y)$ and $\varphi_2(x,y) = \frac{1}{2}(x-y)$. Check: $\varphi_1(1,1) = 1$, $\varphi_1(1,-1) = 0$, $\varphi_2(1,1) = 0$, $\varphi_2(1,-1) = 1$.
3. For $\psi(x,y) = 3x + y$: $\psi = \psi(1,1)\varphi_1 + \psi(1,-1)\varphi_2 = 4\varphi_1 + 2\varphi_2$.



Level lines of the dual basis of $v_1 = (1, 1)$, $v_2 = (1, -1)$. The tip of v_1 lies on $\varphi_1 = 1$ and $\varphi_2 = 0$; the tip of v_2 lies on $\varphi_1 = 0$ and $\varphi_2 = 1$.

The question answered. The space is V', and for $\mathbb{R}^2$ with the basis $(1, 1), (1, -1)$ a basis of it is the dual basis $\varphi_1 = (\frac{1}{2}\ \frac{1}{2})$, $\varphi_2 = (\frac{1}{2}\ -\frac{1}{2})$ of the example.

Shortcuts

- **Equation between functionals: evaluate on a basis.** The j-th coordinate of ψ in the dual basis is $\psi(v_j)$. Reason: HW5 P4(b). Instance: to show $\psi = 2\varphi_1 - \varphi_2$, check both sides at v_1 (value 2) and at v_2 (value -1).
- **Dual basis in $\mathbb{F}^n$: rows of M^{-1}**, where M has the basis vectors as columns. Reason: $M^{-1}M = I$ (the example above). Instance: for the basis $(1, 0), (1, 1)$, $M^{-1} = \left[\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right]$, so the dual basis is $(1\ -1)$, $(0\ 1)$.
- **A nonzero functional is surjective**, so its null space has dimension $\dim V - 1$. Reason: HW5 P3 and rank–nullity. Condition: V finite-dimensional for the dimension count. Instance: $\varphi(x, y, z) = x + y$ on $\mathbb{R}^3$ has a null space of dimension 2.
- **Known dual bases on $\mathcal{P}_n(\mathbb{R})$.** The dual basis of $1, x, \ldots, x^n$ is $\varphi_k(p) = p^{(k)}(0)/k!$, the k-th Taylor coefficient. The functionals $p \mapsto p(c_0), \ldots, p \mapsto p(c_n)$ at distinct points are the dual basis of the Lagrange polynomials $\ell_0, \ldots, \ell_n$ with $\ell_j(c_k) = \delta_{jk}$. Reason: check $\varphi_j(b_k) = \delta_{jk}$ directly. Instance: on $\mathcal{P}_1(\mathbb{R})$ with points $0, 1$, the Lagrange basis is $1 - x$, x (the polynomials of HW5 P1), dual to $p \mapsto p(0)$, $p \mapsto p(1)$.

Traps

- The dual basis depends on the whole basis. For e_1, e_2, $\varphi_1 = (1\ 0)$. For $e_1, (1, 1)$, $\varphi_1 = (1\ -1)$: changing v_2 changed φ_1.
- V' is the dual space, not a derivative, and φ_j is not the j-th vector v_j written as a row: for the basis $(1, 1), (1, -1)$, $\varphi_1 = (\frac{1}{2}\ \frac{1}{2})$, not $(1\ 1)$.

§2.5 The double dual (Ch. 10.1)

The question. There is no preferred isomorphism $V \to V'$. Is there one from V to $V'' = (V')'$? For $V = \mathbb{R}^2$, which element of V'' should correspond to $(3, 5)$?

Definition (Notes 10.1 / LADR 3F: the double dual map)**.** For $v \in V$ let $j_V(v) \in V''$ be "evaluate at v": $j_V(v)(\varphi) = \varphi(v)$ for $\varphi \in V'$. This gives $j_V : V \to V''$.

- $j_V(v)$ is linear in φ: $(\lambda\varphi + \psi)(v) = \lambda\varphi(v) + \psi(v)$. So $j_V(v) \in V''$.
- j_V is linear in v: $\varphi(\lambda v + w) = \lambda\varphi(v) + \varphi(w)$.
- The definition uses no basis and no inner product.

Proposition (Prop. 46 / LADR 3F: $V \to V''$)

j_V is injective. If V is finite-dimensional, j_V is an isomorphism.

The idea

Injective means: if $v \neq 0$ then $j_V(v) \neq 0$, that is, some functional φ has $\varphi(v) \neq 0$. So the task is to build a functional that does not vanish at a given nonzero vector. Functionals are built by prescribing values on a basis, so make v a basis vector and give it the value 1.

Restated: if $\varphi(v) = \varphi(w)$ for every $\varphi \in V'$, then $v = w$. If two vectors have the same value under every functional, they are equal.

Proof. 1. Let $v \neq 0$. The list v is independent; extend it to a basis $v, v_2, \dots$ of V (W1 if V is finite-dimensional; for infinite-dimensional V we cite that every independent list extends to a basis, a consequence of Zorn's lemma that this course does not prove).

2. Let φ be the functional with $\varphi(v) = 1$ and $\varphi(v_k) = 0$ for the other basis vectors.

3. Then $j_V(v)(\varphi) = \varphi(v) = 1 \neq 0$, so $j_V(v) \neq 0$. Hence $\operatorname{null} j_V = \{0\}$.

4. If $\dim V < \infty$: $\dim V'' = \dim V' = \dim V$, and an injective map between spaces of equal finite dimension is an isomorphism (W3).

□

In coordinates. Vectors of $\mathbb{F}^n$ are columns and functionals are rows. A functional on rows is determined by its values on the rows $e_1^{\mathsf{T}}, \dots, e_n^{\mathsf{T}}$, that is, by n numbers, which we write as a column. $j_V(v)$ is the column v itself, since $j_V(v)(e_k^{\mathsf{T}}) = e_k^{\mathsf{T}} v = v_k$.

The question answered. Yes: j_V is defined without choices and is an isomorphism for finite-dimensional V. For $(3,5) \in \mathbb{R}^2$, $j_V(3,5)$ is the function on rows $(a\ b) \mapsto 3a + 5b$; in coordinates it is the column $(3,5)$ again.

Shortcuts

- **To show $v = 0$, find a functional with $\varphi(v) \neq 0$**, or check that every element of a dual basis vanishes at v. Reason: Prop. 46 and HW5 P4(a). Instance: if $\varphi_1(v) = \varphi_2(v) = 0$ for the dual basis of a basis of $\mathbb{R}^2$, then $v = \varphi_1(v)v_1 + \varphi_2(v)v_2 = 0$.

Traps

- $V \cong V'$ in finite dimensions, but not by a preferred map (the example with e_1 above); $V \cong V''$ by the preferred map j_V. In infinite dimensions j_V is injective but need not be surjective (cited; not proved in this course).

§2.6 The dual map and the transpose (Ch. 10.2)

The question. Let $T\colon \mathbb{R}^2 \to \mathbb{R}$, $T(x,y) = 2x - y$ (Practice Midterm 2), and let $\ell(t) = 3t$ be a functional on $\mathbb{R}$. Then $\ell \circ T$ is a functional on $\mathbb{R}^2$. What is it, and what does the map $\ell \mapsto \ell \circ T$ look like as a matrix?

Definition (Def. 28 / LADR 3.118)**.** For $T \in \mathcal{L}(V,W)$ the **dual map** is $T'\colon W' \to V'$, $T'(\varphi) = \varphi \circ T$.

- $T'\varphi = \varphi \circ T$ is a composition of linear maps $V \to W \to \mathbb{F}$, so it is linear: $T'\varphi \in V'$.
- T' is linear: for $\varphi, \psi \in W'$, $\lambda \in \mathbb{F}$ and $v \in V$,

$$(T'(\lambda\varphi + \psi))(v) = (\lambda\varphi + \psi)(Tv) = \lambda\varphi(Tv) + \psi(Tv) = (\lambda T'\varphi + T'\psi)(v).$$

The idea

Why does the direction reverse? Given $\varphi \in W'$ and $T\colon V \to W$, we get a functional on V: first send v to W, then apply φ. That is $\varphi \circ T$. So T' sends functionals on W to functionals on V. There is no way to go the other way: a functional on V gives no values on vectors of W outside the range.

The defining equation to use in every computation: $(T'\varphi)(v) = \varphi(Tv)$.

Example 45 and Practice Midterm 2. (45) $T\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $Tp = xp$, and $\varphi(p) = p(2)$. Then $(T'\varphi)(p) = \varphi(xp) = 2p(2)$, so $T'\varphi = 2\varphi$.

(PM2) $T(x, y) = 2x - y$ from $\mathbb{R}^2$ to $\mathbb{R}$ and $\ell(t) = 3t$. Then $(T'\ell)(x, y) = \ell(2x - y) = 6x - 3y$.

Dual of differentiation (LADR 3.119). Let $D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $Dp = p'$. For $\varphi(p) = p(3)$: $(D'\varphi)(p) = \varphi(p') = p'(3)$. For $\psi(p) = \int_0^1 p$:

$$(D'\psi)(p) = \int_0^1 p'(x)\,dx = p(1) - p(0),$$

so $D'\psi$ is the functional $p \mapsto p(1) - p(0)$.

Proposition (Prop. 47 / LADR 3.120: algebra of dual maps)
The map $\mathcal{L}(V, W) \to \mathcal{L}(W', V')$, $T \mapsto T'$, is linear: $(\lambda T_1 + T_2)' = \lambda T_1' + T_2'$. It reverses products: $(ST)' = T'S'$ for $T \in \mathcal{L}(U, V)$, $S \in \mathcal{L}(V, W)$.

Proof. 1. For $\varphi \in W'$ and $v \in V$:

$$\big((\lambda T_1 + T_2)'\varphi\big)(v) = \varphi(\lambda T_1 v + T_2 v) = \lambda\varphi(T_1 v) + \varphi(T_2 v) = \big(\lambda T_1'\varphi + T_2'\varphi\big)(v).$$

2. Apply the definition of the dual map three times, using that composition is associative:

$$(ST)'\varphi = \varphi \circ (S \circ T) = (\varphi \circ S) \circ T = (S'\varphi) \circ T = T'(S'\varphi).$$

□

Proposition (Prop. 48: j is compatible with every linear map)
For $T\colon V \to W$: $T'' \circ j_V = j_W \circ T$.

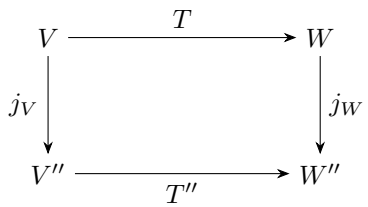
Proof. 1. Fix $v \in V$. Two elements of W'' are equal when they agree at every $\psi \in W'$.

2. For $\psi \in W'$, unwind the definitions one at a time:

$$\big(T''(j_V(v))\big)(\psi) = j_V(v)(T'\psi) = (T'\psi)(v) = \psi(Tv) = j_W(Tv)(\psi).$$

3. So $T''(j_V(v)) = j_W(Tv)$ for every v.

□



Prop. 48: across then down equals down then across.

This is what "canonical" means for j. When V and W are finite-dimensional, j_V and j_W are isomorphisms (Prop. 46), and the equation says $T'' = j_W \circ T \circ j_V^{-1}$: after identifying V with V'' and W with W'', the map T'' is T.

Theorem (Prop. 49 / LADR 3.132: the matrix of the dual map is the transpose)
Let $B = (v_1, \dots, v_n)$, $C = (w_1, \dots, w_m)$ be bases of V, W with dual bases $B' = (v_1', \dots, v_n')$, $C' = (w_1', \dots, w_m')$. If $A = \mathcal{M}(T)$ with respect to B, C, then $\mathcal{M}(T')$ with respect to C', B' is A^{T}, where $(A^{\mathsf{T}})_{ij} = a_{ji}$.

The idea

Column j of $\mathcal{M}(T')$ must hold the coordinates of the functional $T'(w_j') = w_j' \circ T$ in the basis $v_1', \dots, v_n'$. What does $w_j' \circ T$ do? It sends v to the j-th coordinate of Tv. In coordinates, the j-th coordinate of Ax is $a_{j1}x_1 + \cdots + a_{jn}x_n$: the coefficients are *row j of A*. So row j of A becomes column j of $\mathcal{M}(T')$. That is the transpose.

To prove it, find the coordinates of $T'(w_j')$ by evaluating at each v_k (the technique from Prop. 45).

Proof. 1. By definition of A: $Tv_k = \sum_{l=1}^{m} a_{lk} w_l$.

2. Evaluate $T'(w_j')$ at v_k:
$$T'(w_j')(v_k) = w_j'(Tv_k) = w_j'\Big(\sum_l a_{lk} w_l\Big) = a_{jk}.$$

3. The coordinates of a functional ψ in the dual basis are its values $\psi(v_k)$ (HW5 P4(b)). So $T'(w_j') = \sum_{k=1}^{n} a_{jk} v_k'$.

4. So column j of $\mathcal{M}(T')$ is $(a_{j1}, \dots, a_{jn})^{\mathsf{T}}$, which is row j of A.

□

Example 46. Let
$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} : \mathbb{R}^3 \to \mathbb{R}^2.$$
The dual map sends a row $(x\ y)$ to the row $(x\ y)A = (x+4y \ \ 2x+5y \ \ 3x+6y)$. Written as columns of coordinates in the standard dual bases, this is multiplication by
$$A^{\mathsf{T}} = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

Transpose of a product. $(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$, because $(ST)' = T'S'$ (Prop. 47) and the matrix of a dual map is the transpose (Prop. 49).

Distributions and observables. For a Markov chain with stochastic matrix P (W6), let f assign a reward f_i to state i. Under a distribution p the mean reward is $f^{\mathsf{T}}p = \sum_i f_i p_i$, a linear functional of p. After one step the mean reward is
$$f^{\mathsf{T}}(Pp) = (P^{\mathsf{T}}f)^{\mathsf{T}}p, \qquad (P^{\mathsf{T}}f)_j = \sum_i P_{ij} f_i,$$
because $(f^{\mathsf{T}}P)^{\mathsf{T}} = P^{\mathsf{T}}f$ (transpose of a product). So P moves distributions forward, $p \mapsto Pp$, and the dual map moves reward functions, $f \mapsto P^{\mathsf{T}}f$. The entry $(P^{\mathsf{T}}f)_j$ is the average of the rewards f_i weighted by the probabilities P_{ij} of moving from j to i: the mean reward one step later when the present state is j.

Books that write distributions as rows use $A = P^{\mathsf{T}}$: with initial distribution the row λ^{T}, the mean reward after k steps is
$$\lambda^{\mathsf{T}} A^k f = f^{\mathsf{T}} P^k \lambda,$$
the same number in our column convention.

The question answered. $\ell \circ T(x, y) = 6x - 3y$. In the standard bases $\mathcal{M}(T) = \begin{pmatrix} 2 & -1 \end{pmatrix}$, so $\mathcal{M}(T')$ is the column $\left[\begin{smallmatrix} 2 \\ -1 \end{smallmatrix}\right]$ (Prop. 49); ℓ has coordinate 3, and $3\left[\begin{smallmatrix} 2 \\ -1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 6 \\ -3 \end{smallmatrix}\right]$ are the coordinates of $6x - 3y$.

Shortcuts

- **Matrix of T': transpose.** Reason: Prop. 49. Instance: Ex. 46.

- rank $A^\mathsf{T} = \operatorname{rank} A$, so the span of the rows and the span of the columns have the same dimension. Reason: the dimension count after Prop. 51. Instance: $\left[\begin{smallmatrix}1&2&3\\2&4&6\end{smallmatrix}\right]$ has one independent row and one independent column.

Traps

- Orders reverse: $(ST)' = T'S'$ and $(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}$. With $A = \left[\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right]$, $B = \left[\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right]$: $(AB)^\mathsf{T} = \left[\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right]$, but $A^\mathsf{T}B^\mathsf{T} = \left[\begin{smallmatrix}0&0\\0&1\end{smallmatrix}\right]$.
- T' goes from W' to V', opposite to T, and it is not the derivative of T: for $T(x,y) = 2x - y$, $T'\ell = 6x - 3y$ (question above).
- T injective does not make T' injective (Midterm 2 review, true/false). $T\colon \mathbb{R} \to \mathbb{R}^2$, $t \mapsto (t, 0)$ is injective, and T' sends the row $(0\ 1)$ to 0. The correct statement is: T injective $\iff$ T' surjective (Prop. 51).

§2.7 Annihilators (Ch. 10.3)

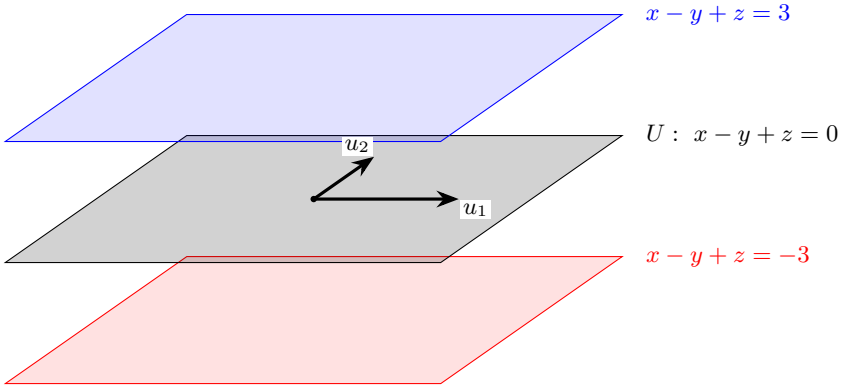
The question. Which rows $(a\ b\ c)$ give 0 on both $(1,1,0)$ and $(0,1,1)$? They form a subspace of $(\mathbb{R}^3)'$; what is its dimension, and how is it related to the null space and range of a map?

Definition (Def. 29 / LADR 3.121)**.** The **annihilator** of a subspace $U \subseteq V$ is

$$U^0 = \{\varphi \in V' : \varphi(u) = 0 \text{ for all } u \in U\}.$$

- U^0 is a subspace of V' (LADR 3.124): the zero functional vanishes on U, and if φ, ψ vanish on U, then $(\lambda\varphi + \psi)(u) = \lambda\varphi(u) + \psi(u) = 0$ for $u \in U$.
- It is enough to check φ on a spanning list $u_1, \dots, u_k$ of U: then $\varphi(\sum a_i u_i) = \sum a_i \varphi(u_i) = 0$.
- $\{0\}^0 = V'$ and $V^0 = \{0\}$.

Example 47. $U = \operatorname{span}\big((1,1,0),(0,1,1)\big)$. A row $(a\ b\ c)$ is in U^0 when $a + b = 0$ and $b + c = 0$, that is, $(a\ b\ c) = b(-1\ \ 1\ \ -1)$. So $U^0 = \operatorname{span}\big((-1\ \ 1\ \ -1)\big) = \operatorname{span}\big((1\ \ -1\ \ 1)\big)$.



The plane $U = \operatorname{span}(u_1, u_2)$, $u_1 = (1,1,0)$, $u_2 = (0,1,1)$, and two other level sets $x - y + z = c$ of the functional $(1\ -1\ 1)$, which spans U^0. The level set $c = 0$ is U itself: the functional vanishes on U.

Proposition (LADR 3.125: dimension of the annihilator)
If V is finite-dimensional, $\dim U^0 = \dim V - \dim U$.

The idea

Membership in U^0 is a condition on the values $\psi(u_1), \dots, \psi(u_k)$. In a dual basis those values are exactly the first k coordinates of ψ (HW5 P4(b)). So choose a basis of V that starts with a basis of U: then $\psi \in U^0$ says "the first k coordinates are 0", and the remaining $n-k$ dual vectors form a basis of U^0. Midterm 2 Q1 (R3) is this construction with numbers.

Proof. 1. Let $u_1, \dots, u_k$ be a basis of U; extend to a basis $u_1, \dots, u_k, x_1, \dots, x_r$ of V, with dual basis $\varphi_1, \dots, \varphi_k, \chi_1, \dots, \chi_r$.

2. Any $\psi \in V'$ is $\psi = \sum_i \psi(u_i)\varphi_i + \sum_j \psi(x_j)\chi_j$ (coordinates of a functional).

3. $\psi \in U^0$ exactly when $\psi(u_i) = 0$ for all i, that is, exactly when $\psi \in \operatorname{span}(\chi_1, \dots, \chi_r)$.

4. $\chi_1, \dots, \chi_r$ is independent (part of a basis), so it is a basis of U^0, and $r = \dim V - \dim U$.

□

Proposition (LADR 3.127: when the annihilator is everything or nothing)
Let V be finite-dimensional and $U \subseteq V$ a subspace. Then $U^0 = V' \iff U = \{0\}$, and $U^0 = \{0\} \iff U = V$.

Proof. 1. By LADR 3.125, $\dim U^0 = \dim V - \dim U$.

2. $U^0 = V'$ exactly when $\dim U^0 = \dim V' = \dim V$ (W1, a subspace of full dimension), that is, when $\dim U = 0$, that is, $U = \{0\}$.

3. $U^0 = \{0\}$ exactly when $\dim U^0 = 0$, that is, $\dim U = \dim V$, that is, $U = V$ (W1).

□

Theorem (Prop. 50 / LADR 3.128, 3.130: null space and range of the dual map)
For $T \in \mathcal{L}(V, W)$: (i) $\operatorname{null} T' = (\operatorname{range} T)^0$; (ii) $\operatorname{range} T' = (\operatorname{null} T)^0$.

The idea

(i) is a translation. $T'\varphi = 0$ means $\varphi(Tv) = 0$ for all v, which says φ vanishes on every vector of the form Tv, that is, on $\operatorname{range} T$.

(ii) has an easy half and a half that needs a construction. Easy: a functional $\psi \circ T$ vanishes wherever T does. Harder: given $\varphi \in V'$ that vanishes on $\operatorname{null} T$, we must produce $\psi \in W'$ with $\psi \circ T = \varphi$. The equation tells us what ψ has to be on $\operatorname{range} T$: $\psi(Tv) = \varphi(v)$. Is that a valid definition? The same vector of $\operatorname{range} T$ can be Tv and Tw; then $v - w \in \operatorname{null} T$, and $\varphi(v) = \varphi(w)$ because φ vanishes on $\operatorname{null} T$. So the hypothesis is exactly the well-definedness condition. Outside $\operatorname{range} T$ the equation imposes nothing, so extend ψ in any way.

Proof. 1. **(i).** For $\varphi \in W'$:

$$\varphi \in \operatorname{null} T' \iff \varphi \circ T = 0 \iff \varphi(Tv) = 0 \text{ for all } v \in V \iff \varphi \in (\operatorname{range} T)^0.$$

2. **(ii), $\operatorname{range} T' \subseteq (\operatorname{null} T)^0$.** Assume $\varphi \in \operatorname{range} T'$, so $\varphi = T'\psi = \psi \circ T$ for some $\psi \in W'$. For $v \in \operatorname{null} T$: $\varphi(v) = \psi(Tv) = \psi(0) = 0$. So $\varphi \in (\operatorname{null} T)^0$.

3. **(ii), $(\operatorname{null} T)^0 \subseteq \operatorname{range} T'$.** Assume φ vanishes on $\operatorname{null} T$. Define $\psi_0 \colon \operatorname{range} T \to \mathbb{F}$ by $\psi_0(Tv) = \varphi(v)$. It is well defined: if $Tv = Tw$, then $v - w \in \operatorname{null} T$, so $\varphi(v) - \varphi(w) = \varphi(v - w) = 0$. It is linear:

$$\psi_0(\lambda Tv + Tw) = \psi_0(T(\lambda v + w)) = \lambda\varphi(v) + \varphi(w) = \lambda\psi_0(Tv) + \psi_0(Tw).$$

4. Extend a basis of $\operatorname{range} T$ to a basis of W, and let $\psi \in W'$ equal ψ_0 on the basis vectors of $\operatorname{range} T$ and 0 on the added ones. Then $\psi = \psi_0$ on $\operatorname{range} T$. (If W is infinite-dimensional, this uses the cited fact from Prop. 46, step 1.)

5. $T'\psi = \psi \circ T$ sends v to $\psi(Tv) = \psi_0(Tv) = \varphi(v)$. So $T'\psi = \varphi$ and $\varphi \in \operatorname{range} T'$.

□

Proposition (Prop. 51 / LADR 3.129, 3.131)

Let V, W be finite-dimensional and $T \in \mathcal{L}(V, W)$. Then T is injective $\iff$ T' is surjective, and T is surjective $\iff$ T' is injective.

The idea

Prop. 50 turns statements about T' into statements about annihilators. T injective means $\operatorname{null} T = \{0\}$; its annihilator is all of V', which is $\operatorname{range} T'$. T surjective means $\operatorname{range} T = W$; its annihilator is $\{0\}$, which is $\operatorname{null} T'$. The only thing to check is that taking annihilators loses nothing: $U^0 = V'$ only for $U = \{0\}$, and $U^0 = \{0\}$ only for $U = V$ (LADR 3.127).

Proof. 1. The chain for injectivity, using LADR 3.127 in V and Prop. 50(ii):

$$T \text{ injective} \iff \operatorname{null} T = \{0\} \iff (\operatorname{null} T)^0 = V' \iff \operatorname{range} T' = V'.$$

2. The chain for surjectivity, using LADR 3.127 in W and Prop. 50(i):

$$T \text{ surjective} \iff \operatorname{range} T = W \iff (\operatorname{range} T)^0 = \{0\} \iff \operatorname{null} T' = \{0\}.$$

□

The direction "T surjective $\Rightarrow$ T' injective" needs no finite dimension: R2(c).

Dimensions (finite dimensions):

$$\begin{aligned}\dim \operatorname{range} T' &= \dim (\operatorname{null} T)^0 = \dim V - \dim \operatorname{null} T = \dim \operatorname{range} T,\\ \dim \operatorname{null} T' &= \dim (\operatorname{range} T)^0 = \dim W - \dim \operatorname{range} T = \dim W - \dim V + \dim \operatorname{null} T.\end{aligned}$$

For a matrix A: the range of A is spanned by its columns. By Prop. 49, A^{T} is the matrix of T', whose range is spanned by the columns of A^{T}, that is, the rows of A. So the span of the rows and the span of the columns have the same dimension: $\operatorname{rank} A^{\mathsf{T}} = \operatorname{rank} A$.

Proposition (Midterm 2 review: duals of the inclusion and of the quotient map)

Let $U \subseteq V$, V finite-dimensional, $i\colon U \to V$ the inclusion and $\pi\colon V \to V/U$ the quotient map. Then (i) $i'\colon V' \to U'$ is restriction, $\varphi \mapsto \varphi|_U$; it is surjective with null space U^0, so $V'/U^0 \cong U'$; (ii) $\pi'\colon (V/U)' \to V'$ is injective with range U^0, so $(V/U)' \cong U^0$.

Proof. 1. $i'\varphi = \varphi \circ i$ sends u to $\varphi(u)$, so it is $\varphi|_U$.

2. i is injective, so i' is surjective (Prop. 51). $\operatorname{null} i' = (\operatorname{range} i)^0 = U^0$ (Prop. 50(i)). The first isomorphism theorem gives $V'/U^0 \cong U'$.

3. π is surjective, so π' is injective (Prop. 51). $\operatorname{range} \pi' = (\operatorname{null} \pi)^0 = U^0$ (Prop. 50(ii)). So π' is an isomorphism from $(V/U)'$ onto U^0.

□

This explains why $\dim U^0 = \dim V - \dim U = \dim(V/U)$.

The question answered. The rows are the multiples of $(1\ -1\ 1)$ (Ex. 47), a subspace of dimension $1 = 3 - 2$ (LADR 3.125). If $T\colon \mathbb{R}^2 \to \mathbb{R}^3$ has $\operatorname{range} T = U$ (for example the matrix with columns $(1, 1, 0), (0, 1, 1)$), this subspace is $\operatorname{null} T'$ (Prop. 50(i)).

Shortcuts

- **Annihilator in $\mathbb{F}^n$: a linear system.** Solve $(\text{row})\cdot u = 0$ for each spanning vector u of U; expect dimension $n - \dim U$. Reason: the bullets after Def. 29 and LADR 3.125. Instance: Ex. 47.
- **Basis of U^0 from a dual basis.** Extend a basis $u_1, \dots, u_k$ of U to a basis of V; the last $n-k$ dual-basis functionals are a basis of U^0. Reason: proof of LADR 3.125. Instance: Midterm 2 Q1 (R3): for $B = (1,1,0), (0,1,1), (1,0,1)$, the third dual vector $\frac{1}{2}(1\ -1\ 1)$ spans U^0.
- **Annihilators reverse inclusions.** $U \subseteq W \Rightarrow W^0 \subseteq U^0$, and $(U+W)^0 = U^0 \cap W^0$. Reason: a functional vanishes on $U+W$ exactly when it vanishes on U and on W. Condition: none. Instance: in $\mathbb{R}^3$, $(\operatorname{span}(e_1) + \operatorname{span}(e_2))^0$ is the set of rows $(0\ 0\ c)$.
- **Injective and surjective swap under the dual.** $\operatorname{null} T' = (\operatorname{range} T)^0$, $\operatorname{range} T' = (\operatorname{null} T)^0$. Reason: Props. 50, 51. Instance: $T\colon \mathbb{R}^2 \to \mathbb{R}^3$, $T(x,y) = (x,y,0)$ is injective, so T' is onto $(\mathbb{R}^2)'$; T is not surjective, so T' is not injective: T' sends the row $(0\ 0\ 1)$ to 0.

Traps

- $U^0 \subseteq V'$, not V. For $U = \operatorname{span}((1,1,0))$, U^0 is a set of rows (functionals), for example $(1\ -1\ 0)$; it is not the plane of vectors perpendicular to U, which is a description that needs the dot product.
- U^0 depends on the space V around U, not only on U: the annihilator of $\operatorname{span}(e_1)$ in $(\mathbb{R}^2)'$ is $\operatorname{span}((0\ 1))$, of dimension 1, while in $(\mathbb{R}^3)'$ it is $\operatorname{span}((0\ 1\ 0), (0\ 0\ 1))$, of dimension 2.

§3 Problem set

HW5 P1. Define $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathbb{R}^2$, $T(p) = (p(0), p(1))$. (a) Determine $\operatorname{null} T$ and prove that T is surjective. (b) Describe explicitly the induced isomorphism $\mathcal{P}_3(\mathbb{R})/\operatorname{null} T \to \mathbb{R}^2$ from the first isomorphism theorem. (c) Find a basis of $\mathcal{P}_3(\mathbb{R})/\operatorname{null} T$.

Hint: a polynomial with zeros at 0 and 1 is divisible by $x(x-1)$ (W7). For (c), an isomorphism carries a basis back to a basis.

HW5 P2. Let $T\colon V \to W$ be linear and U a subspace of V contained in $\operatorname{null} T$. Prove that there is a unique linear map $\widetilde{T}\colon V/U \to W$ with $\widetilde{T}(v+U) = Tv$ for every v. Determine $\operatorname{null} \widetilde{T}$ in terms of $\operatorname{null} T$ and U.

Hint: well-definedness is the only issue for existence.

HW5 P3 (LADR 3F, Exercise 1). Prove that every linear functional $\varphi\colon V \to \mathbb{F}$ is either the zero map or surjective.

Hint: scale a vector with $\varphi(v) \neq 0$.

HW5 P4. Let $v_1, \dots, v_n$ be a basis of V with dual basis $\varphi_1, \dots, \varphi_n$. (a) Prove $v = \sum_{j=1}^{n} \varphi_j(v) v_j$ for every $v \in V$. (b) Prove $\psi = \sum_{j=1}^{n} \psi(v_j)\varphi_j$ for every $\psi \in V'$.

Hint: (a) apply φ_j to $v = \sum a_k v_k$. (b) evaluate both sides at v_k.

HW5 P5. Let $T\colon V \to W$ be linear between finite-dimensional spaces with matrix A in bases B, C. Prove that the matrix of $T'\colon W' \to V'$, $T'(\psi) = \psi \circ T$, in the dual bases is A^{T}.

Hint: evaluate $T'(w_j')$ at v_k.

HW5 P6. Let $U = \operatorname{span}\big((1,1,0), (0,1,1)\big) \subseteq \mathbb{R}^3$. (a) Determine $U^0 \subseteq (\mathbb{R}^3)'$, writing functionals as row vectors. (b) Verify that $\dim U + \dim U^0 = 3$.

Hint: a row is in U^0 when it gives 0 on both spanning vectors (Ex. 47); for (b) use $\dim U^0 = \dim V - \dim U$.

HW5 P7 (optional). Let $j_V\colon V \to V''$, $j_V(v)(\varphi) = \varphi(v)$. Prove that j_V is injective, and an isomorphism when V is finite-dimensional.

Hint: for $v \neq 0$ make v a basis vector and take the functional that is 1 on v and 0 on the others (Prop. 46).

R1 (Practice Midterm 2, Question 1). Let $V = \mathcal{P}_3(\mathbb{R})$, $E\colon V \to \mathbb{R}^2$, $E(p) = (p(0), p(1))$, $U = \operatorname{null} E$. (a) Prove that E is linear and surjective. (b) Find a basis of U. (c) Prove that $\widetilde{E}(p + U) = E(p)$ defines a linear isomorphism $V/U \to \mathbb{R}^2$. (d) Give a basis of V/U and the images of its elements under $\widetilde{E}$.

Hint: same map as HW5 P1; (c) is the first isomorphism theorem, checked by hand.

R2 (Practice Midterm 2, Question 2). Let $T\colon V \to W$, $S\colon W \to Z$ be linear maps between finite-dimensional real spaces, with dual maps $T^*\colon W^* \to V^*$, $S^*\colon Z^* \to W^*$. (a) Give the definition of T^* and prove it is linear. (b) Prove $(S \circ T)^* = T^* \circ S^*$. (c) Prove that if T is surjective then T^* is injective. (d) For $T\colon \mathbb{R}^2 \to \mathbb{R}$, $T(x, y) = 2x - y$ and $\ell(t) = 3t$, compute $T^*\ell$.

Hint: everything follows from $(T^*\varphi)(v) = \varphi(Tv)$; for (c), $T^*\varphi = 0$ means φ vanishes on $\operatorname{range} T = W$.

R3 (Midterm 2, Question 1). Let $V = \mathbb{R}^3$ and $U = \operatorname{span}\big((1, 1, 0), (0, 1, 1)\big)$. (a) What is meant by U^0, the annihilator of U in V'? (b) Find a basis of U^0. (c) Extend the two vectors to a basis B of V and find its dual basis. (d) For the basis B chosen in (c), explain why the third element of its dual basis spans U^0.

Hint: (c) put B as the columns of M; the dual basis is the rows of M^{-1}. (d) the third dual vector vanishes on the first two basis vectors, which span U; then count dimensions with LADR 3.125.

R4 (Midterm 2 review, Problem 1). Let $V = \operatorname{span}(1, \sin x, \sin 2x)$, real functions on $[-\pi, \pi]$. (a) Show that $\delta_0(f) = f(0)$, $\delta_{\pi/2}(f) = f(\pi/2)$, $\delta_0'(f) = f'(0)$ form a basis of V'. (b) Write the dual basis $\varphi_1, \varphi_2, \varphi_3$ of $1, \sin x, \sin 2x$ as combinations of $\delta_0, \delta_{\pi/2}, \delta_0'$. (c) For the functional $\varphi(f) = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx$, write φ in the basis $\varphi_1, \varphi_2, \varphi_3$ and then in $\delta_0, \delta_{\pi/2}, \delta_0'$.

Hint: record the values $\delta_i(b_j)$ in a 3×3 matrix; three functionals in a 3-dimensional V' are a basis when that matrix is invertible, and its inverse gives the dual basis. For (c) use HW5 P4(b).

R5 (Midterm 2 review, Problem 2). Let $U \subseteq V$ with V finite-dimensional, $i\colon U \to V$ the inclusion and $\pi\colon V \to V/U$ the quotient map. (a) Describe i' and show it is surjective with null space U^0. (b) Show π' is injective with range U^0. (c) Conclude $V'/U^0 \cong U'$ and $(V/U)' \cong U^0$.

Hint: Prop. 50 and Prop. 51 applied to i and π; then the first isomorphism theorem.

R6 (Midterm 2 review, true or false). Decide, with a proof or a counterexample: (a) if $T \in \mathcal{L}(V, W)$ is injective, then T' is injective; (b) if T is surjective, then T' is injective; (c) for finite-dimensional V and a subspace U, $\dim U^0 = \dim(V/U)$.

Hint: (a) try $T\colon \mathbb{R} \to \mathbb{R}^2$, $t \mapsto (t, 0)$; (b) Prop. 51 or R2(c); (c) LADR 3.125 and Prop. 40.

E1 (dual basis computation). Find the dual basis of the basis $v_1 = (2, 1)$, $v_2 = (1, 1)$ of $\mathbb{R}^2$, as row vectors. Then write the functional $\psi(x, y) = 3x + y$ in that dual basis.

Hint: the dual basis is the rows of M^{-1} for M with columns v_1, v_2; the coordinates of ψ are $\psi(v_1), \psi(v_2)$.